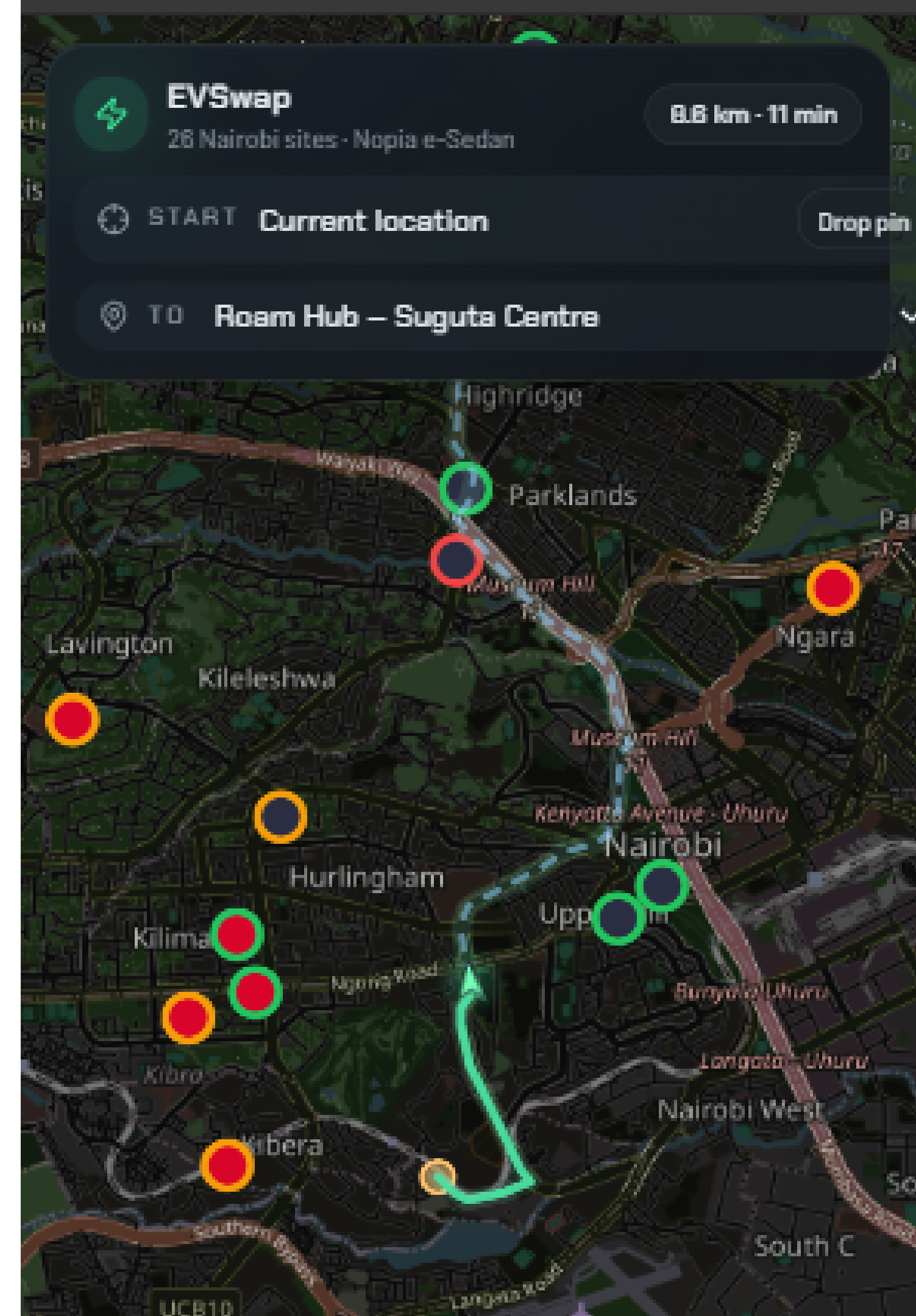


BUILT IN NAIROBI, BUILT FOR THE WORLD



Leveraging AI and City Data to Accelerate Urban E-mobility

AI X CITY HACKATHON 2026



Meet Halima . . .

2 Hours

Daily Commute to School

- Noise
- Emissions
- Fatigue



Photo by Mary Taylor

02 | THE PROBLEM

Kenya's transport sector emits 11.25M tonnes CO₂e a year.

- ⚡ E-mobility (buses, e-bikes, boda-bodas) is the fastest mitigation lever available today.
- 📍 Infrastructure is currently sited by **commercial logic**, not transit need.
- ⚠️ Adoption stalls wherever access lags, creating a cycle of exclusion.

TRANSPORT SECTOR
CONTRIBUTION

MAJOR

EMISSIONS SOURCE

03 | OUR SOLUTION

An AI-driven Model using City Data to Accelerate Urban E-mobility.

We score every part of Nairobi for charging need using real population + transit data, recommending phased sites per the **Avoid-Shift-Improve** framework.



DATA-LED SITING

Leveraging real-world population and transit demand data to identify high-impact locations.



PHASED ROLLOUT

Strategic prioritization: Electrifying buses first, followed by e-bikes and boda-bodas.

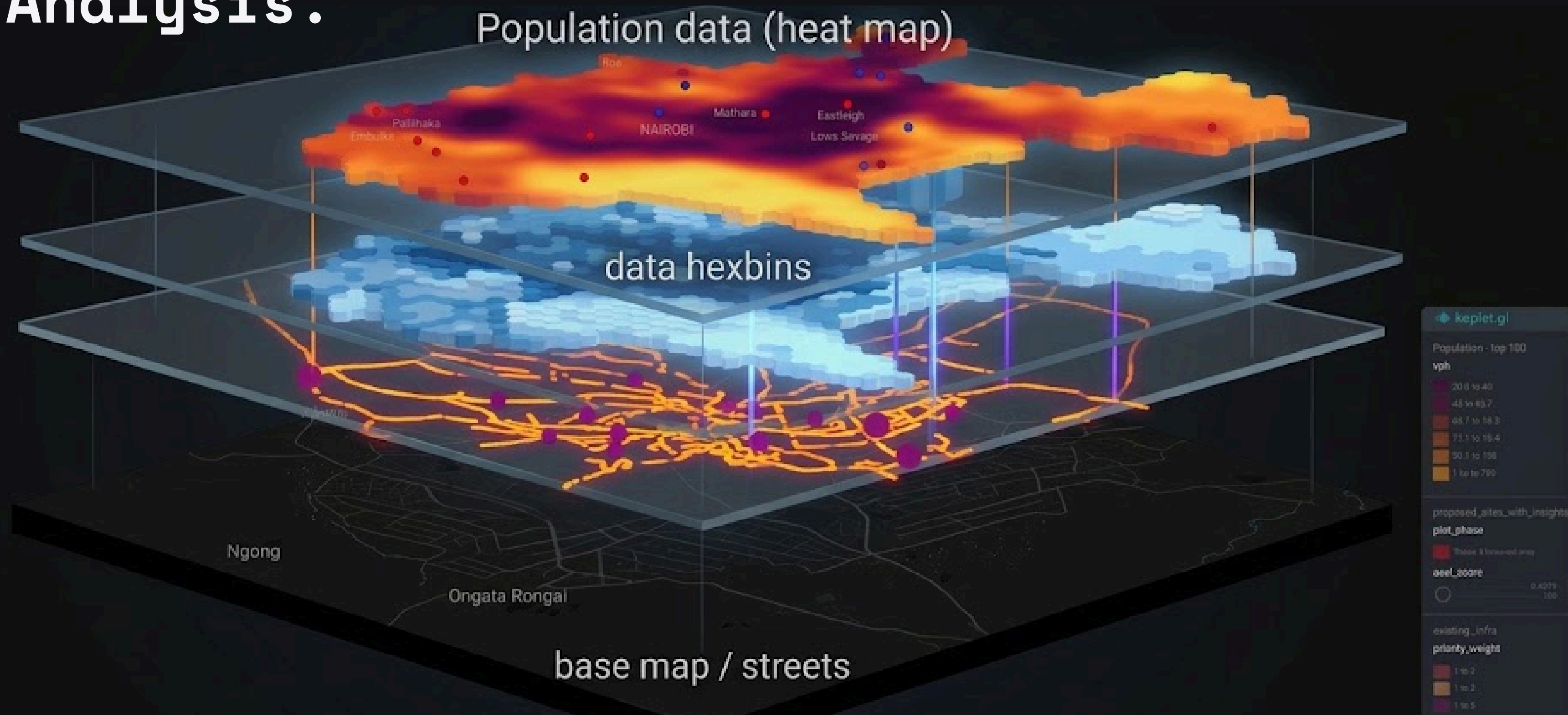


GLOBAL REPLICABILITY

Open data pipelines that can be deployed in any urbanizing city.

04 | METHODOLOGY

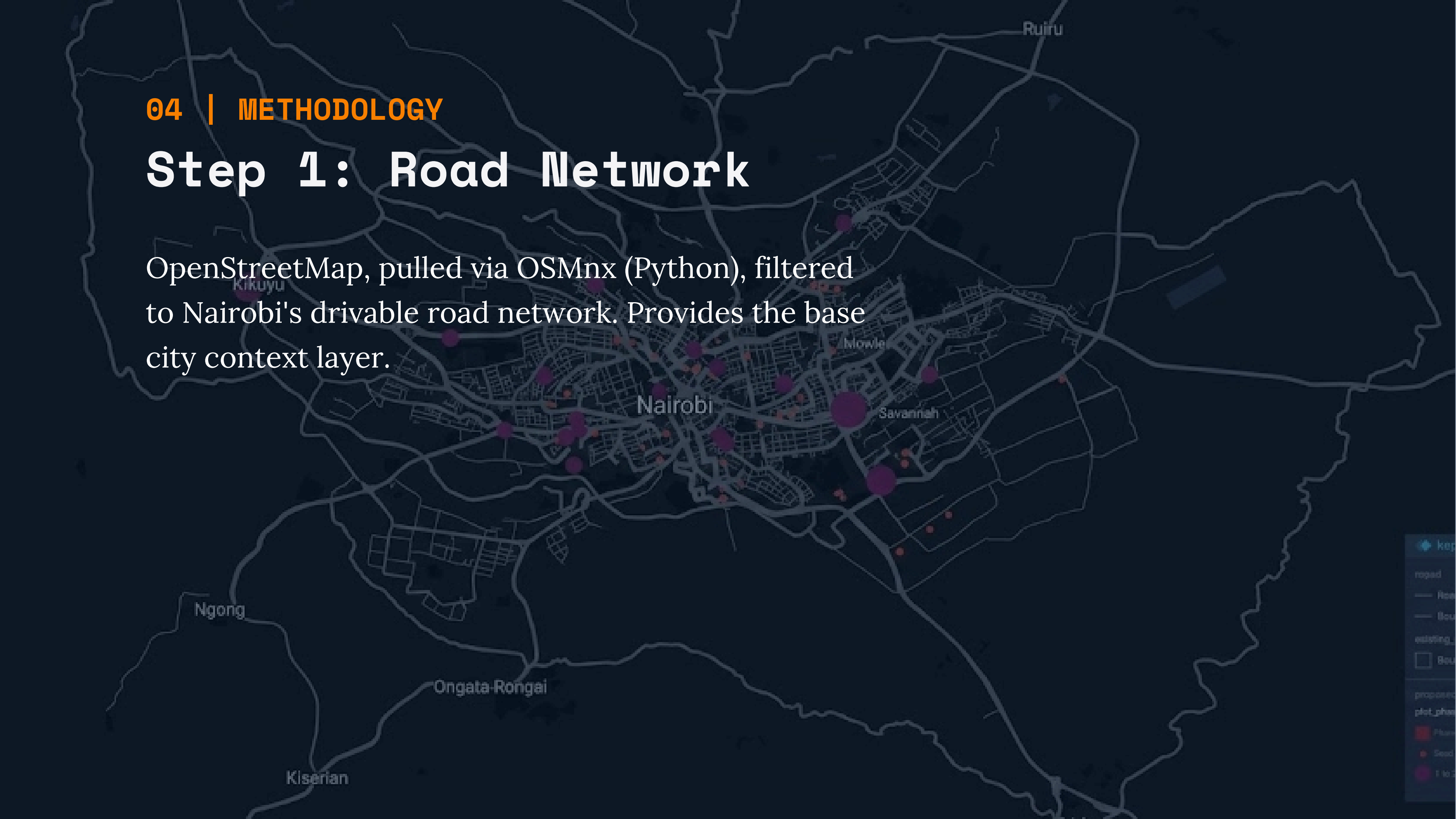
Four Step Spatial Methodology for City Analysis.



04 | METHODOLOGY

Step 1: Road Network

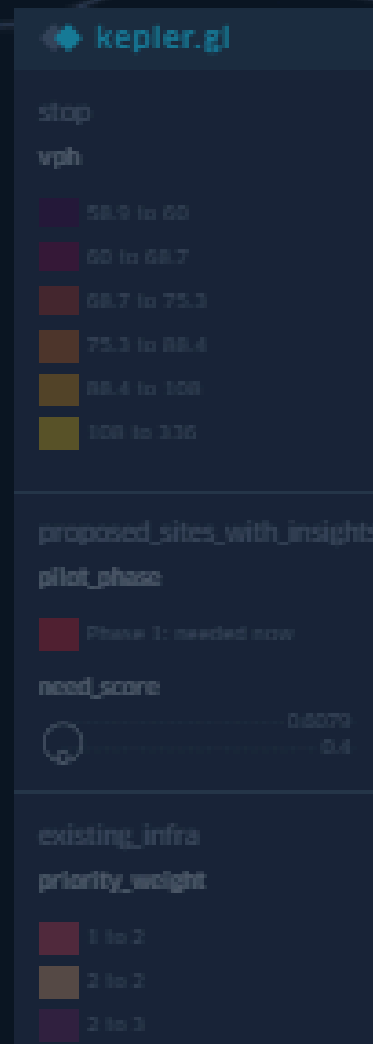
OpenStreetMap, pulled via OSMnx (Python), filtered to Nairobi's drivable road network. Provides the base city context layer.

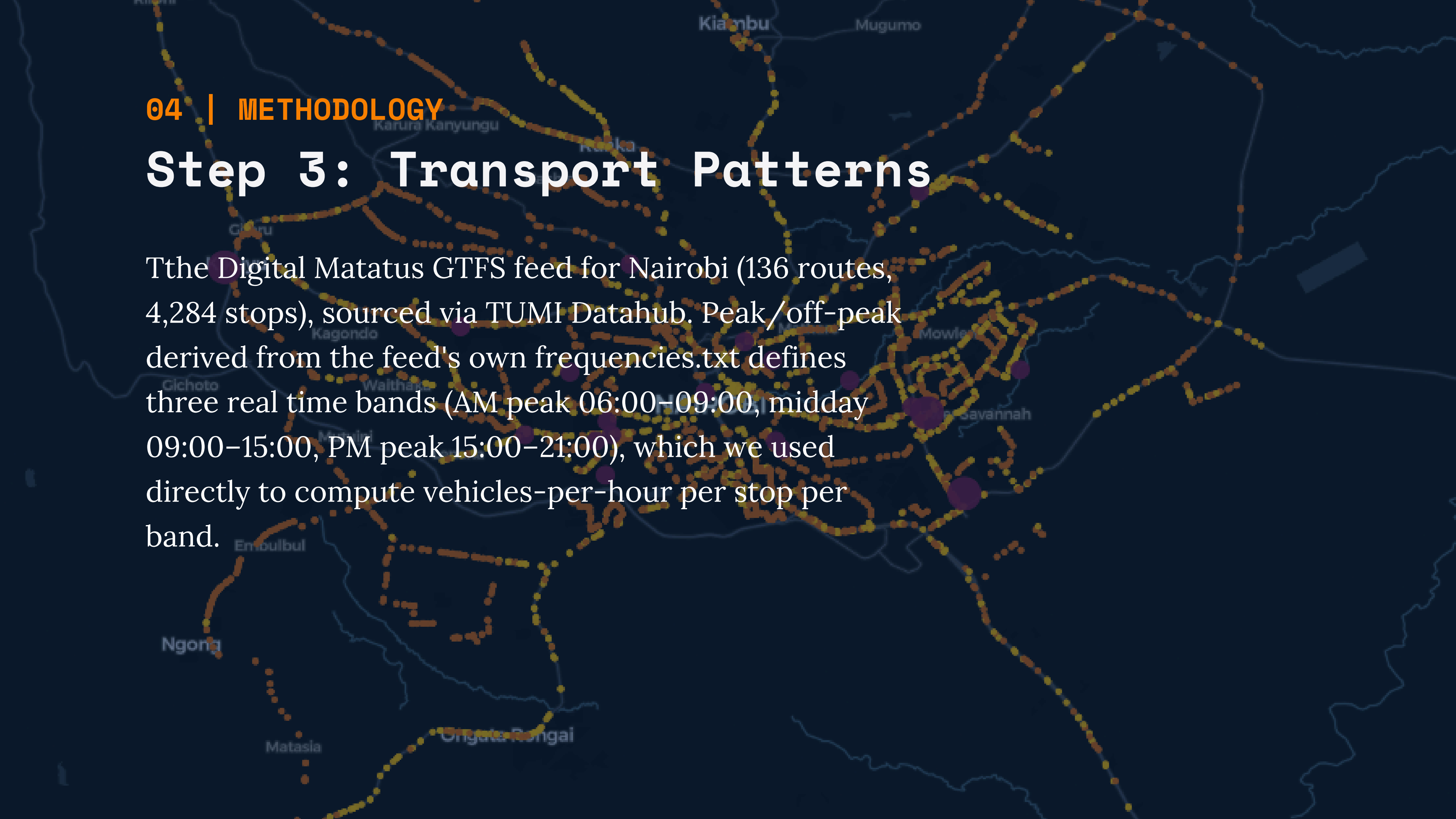


04 | METHODOLOGY

Step 2: Population Data

WorldPop's 100m-resolution, constrained, UN-adjusted gridded population estimate for Kenya (2020). "Constrained" means it estimates population only where actual building footprints exist. Zonal-summed into each hexagon. (Total came out to 4.62M; consistent with Nairobi's known population.)



A map of Nairobi, Kenya, showing a dense network of bus routes. The routes are represented by yellow and orange lines with dots at stops. The map is dark blue with white text. The title 'Step 3: Transport Patterns' is prominently displayed in the upper left. The background map shows various districts and landmarks, with labels like 'Kiaambu', 'Mugumo', 'Karura Kanyungu', 'Kagondo', 'Waltham', 'Mowla', 'Savannah', 'Embulbul', 'Ngong', 'Matasia', and 'Ongata Rongai'.

04 | METHODOLOGY

Step 3: Transport Patterns

The Digital Matatus GTFS feed for Nairobi (136 routes, 4,284 stops), sourced via TUMI Datahub. Peak/off-peak derived from the feed's own frequencies.txt defines three real time bands (AM peak 06:00–09:00, midday 09:00–15:00, PM peak 15:00–21:00), which we used directly to compute vehicles-per-hour per stop per band.

04 | METHODOLOGY

Step 4: Multi Criteria Need-analysis

A composite model per hexagon: 60% population density + 40% peak transit demand (both min-max normalized), discounted by existing e-mobility charging/swap coverage.



Multi-Criteria Need-Analysis Model

```
demand_score = 0.6 × pop_norm + 0.4 ×  
transit_norm
```

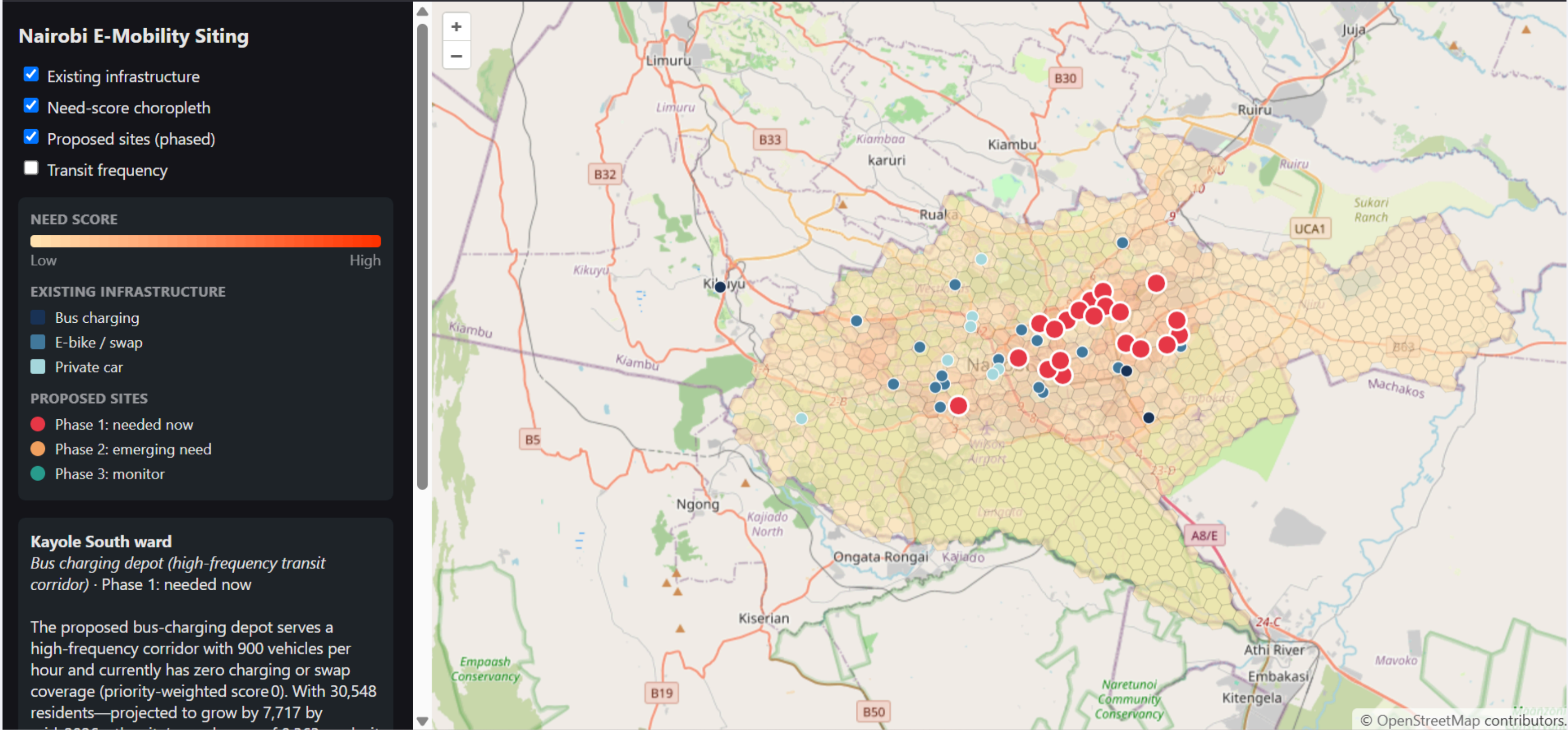
```
existing_coverage = ∑ priority_weight  
(point stations in hex)
```

```
need_score = demand_score × (1 -  
coverage_norm)
```

865

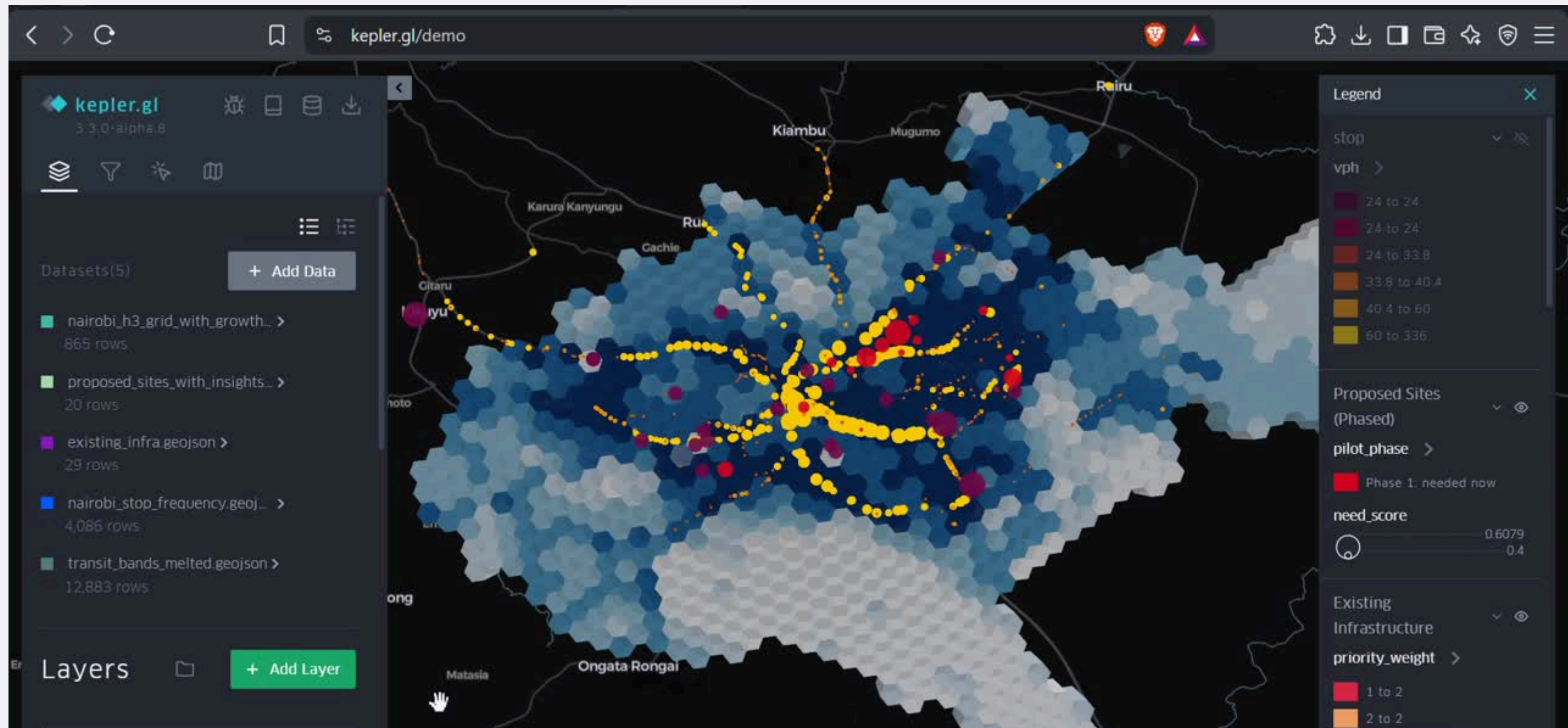
hexagons ranked by
e-mobility infrastructure need

Proposed Sites + Live AI Insights



08 | LIVE DEMO PREVIEW

Fully interactive insights Dashboard.

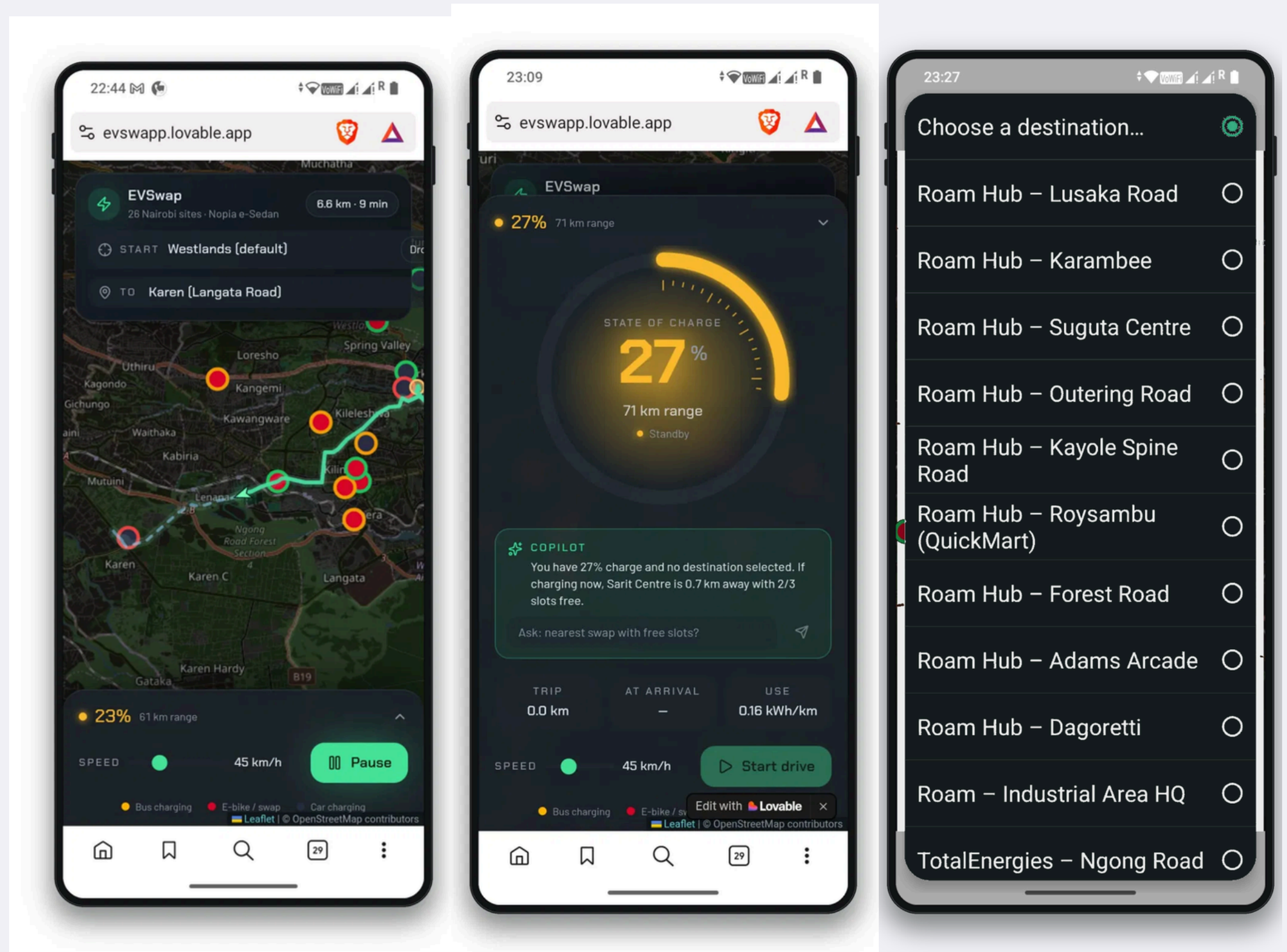


Designed for decisions, not just discovery.

LIVE PROTOTYPE AVAILABLE

09 | LIVE DEMO PREVIEW

Mobile-First EVSwapp App



Designed for decisions, not just discovery.

evswapp.lovable.app

11 | BUSINESS MODEL & IMPACT

Here is How we Make Money

SITING-AS-A-SERVICE

Charge-point operators pay for verified, ranked, AI-explained site recommendations, turning high-risk capex into data.

MODEL: PER-REPORT / SUB

REAL ESTATE MATCHMAKING

Property owners list sites; we score them against demand and match them to operators for a conversion fee.

MODEL: BROKERAGE FEE

ADVERTISING INTELLIGENCE

High-need sites are high-traffic sites. We license footfall data and take a share of station digital-screen ad revenue.

MODEL: DATA LICENSE / REV SHARE

TARGET POPULATION

Underserved Phase-1 Hotspots

MODE SHIFT

Petrol Bodaas → E-Bikes & Buses

ALIGNMENT

C40 Transport-Emissions Goals

"Every stream monetizes the same core asset – the need-score data – sold to a different buyer for a different decision."

12 | OUR ASK & THE TEAM

We Request Support to Develop and Pilot EVSwapp With a Matched City.



Priscila Adhiambo

GIS Engineer | Urban Planner



Fauzia Omala

Software Engineer



Benard Kimani

Software Engineer



Keziah Nakhanu

Business Lead



Chibuike Cletus

AI/ML Engineer



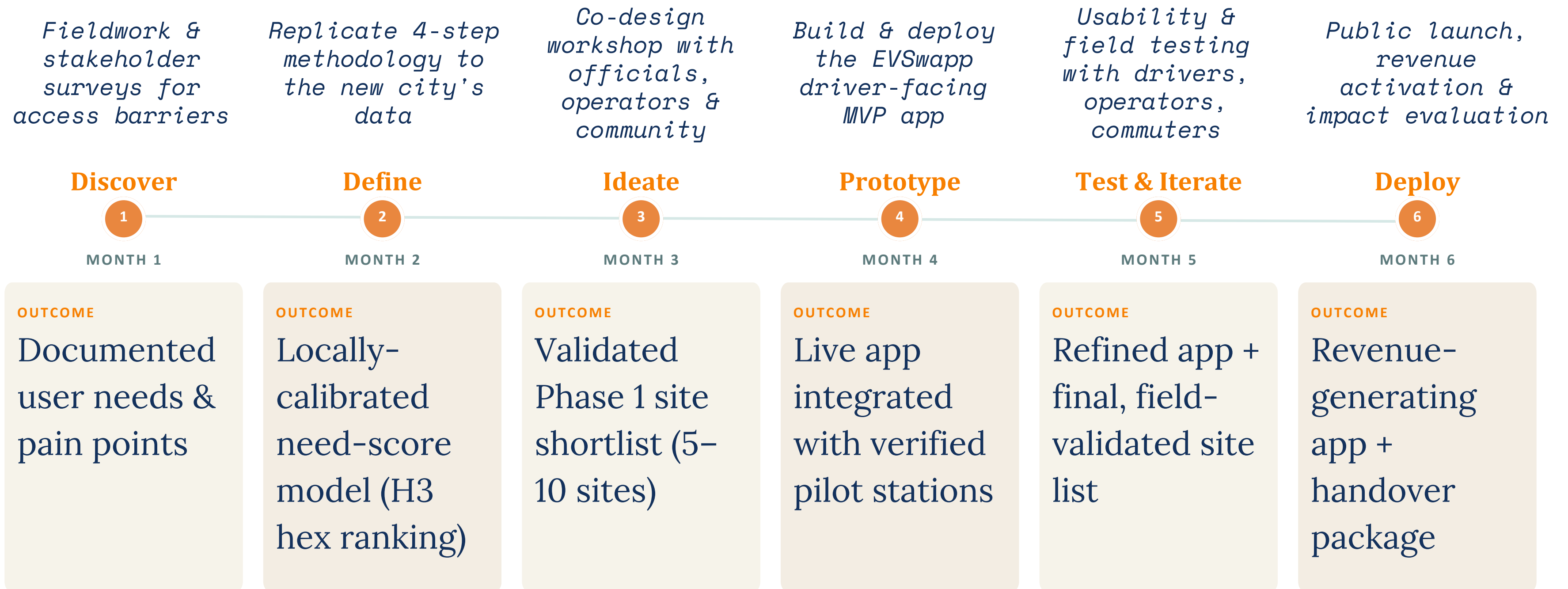
Lynne Adhiambo

Urban Designer

Built in Nairobi, Built for the World.

12 | EVSwapp Pilot Plan

A 6-month, human-centered design process



Let's
Secure
Halima's
Future.



Photo by Mary Taylor